Strand 3: Geometry and Measurement
Sub-strand 1: Shapes and Space

Content Standard: B9.3.1.1
Indicator: B9.3.1.1.1

Lesson Topic: Polygons, Angle Properties, and the Derivation of Angle Formulas
Core Mandate: First-principles derivation, geometric proofs, diagrammatic modeling, and algebraic solutions.

1. Conceptual Foundations and Terminology Breakdown

To build genuine geometric intuition, we examine terms from their mathematical origins. The word **polygon** stems from Greek *polus* ('many') and *gonia* ('angle/corner'). A polygon is a closed two-dimensional plane figure bounded strictly by straight line segments joined consecutively end-to-end. The bounding line segments are called **sides** or **edges**, and the sharp meeting points are called **vertices** (singular: *vertex*).

Convex vs Concave Polygons:

- **Convex Polygon:** Every interior angle measures strictly less than 180°. Every vertex points outward. Any straight line drawn between any two points inside the polygon lies entirely inside the figure. *All standard syllabus theorems apply to convex polygons.*
- **Concave Polygon:** Contains at least one reflex interior angle (between 180° and 360°), forming an internal 'cave' where at least one vertex points inward.

Regular vs Irregular Polygons:

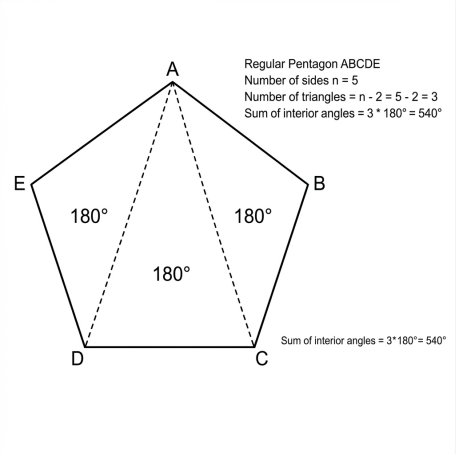
- **Regular Polygon:** A polygon that is simultaneously **equilateral** (all sides of equal length) and **equiangular** (all interior angles of equal magnitude), such as an equilateral triangle or square.
- **Irregular Polygon:** A polygon whose sides and/or interior angles vary in measure.

Systematic Polygon Nomenclature (3 to 12 Sides):

Sides (n)	Name of Polygon	Number of Diagonals (n - 3)	Number of Triangles (n - 2)	Interior Angle Sum
3	Triangle (Trigon)	0	1	180°
4	Quadrilateral (Tetragon)	1	2	360°
5	Pentagon	2	3	540°
6	Hexagon	3	4	720°
7	Heptagon (Septagon)	4	5	900°
8	Octagon	5	6	1080°
9	Nonagon (Enneagon)	6	7	1260°
10	Decagon	7	8	1440°
11	Undecagon (Hendecagon)	8	9	1620°
12	Dodecagon	9	10	1800°
n	n-gon	n - 3	n - 2	(n - 2) × 180°

2. First-Principles Derivations of Geometric Angle Formulas

Rather than accepting formulas without proof, we use the geometric principle of **triangulation**. From any single anchor vertex A in a convex polygon of n sides, we draw straight diagonals to all other non-adjacent vertices.



The Triangulation Proof:

1. From anchor vertex A, you cannot draw diagonals to A itself or to its 2 adjacent neighbor vertices (which are exterior sides).
   Diagonals from 1 vertex = n - 3

2. These diagonals divide the polygon into non-overlapping triangles:
   Number of triangles = (n - 3) + 1 = n - 2

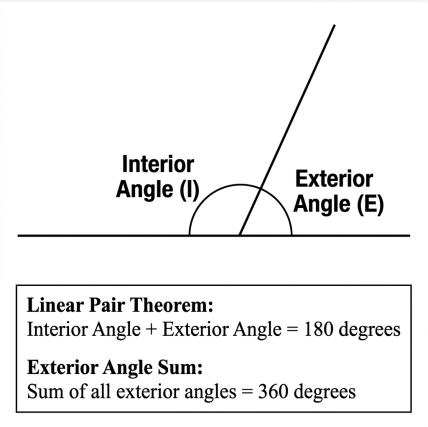
3. Each triangle possesses an angle sum of exactly 180°. Hence:
   Sum of Interior Angles S = (n - 2) × 180°

4. For a regular polygon (equal angles):
   Each Interior Angle = [(n - 2) × 180°] / n

Figure 1.1: Triangulation Proof of the Interior Angle Sum Theorem (Pentagon n = 5, n - 2 = 3 triangles).

3. Exterior Angles and the Perimeter Tour Theorem

An **exterior angle** is formed when any side of a polygon is extended past a vertex. Because the interior angle and exterior angle sit together on a straight line, they form a linear pair.



Key Exterior Angle Theorems:

1. Linear Pair Identity:
Interior Angle + Exterior Angle = 180°

2. Exterior Angle Sum Theorem:
For *any* convex polygon with n sides:
Sum of all Exterior Angles = 360°
Physical Intuition: Walking around any closed perimeter makes one complete 360° revolution.

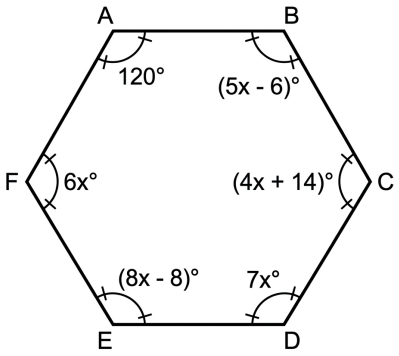
3. Regular Polygon Dimensions:
Each Exterior Angle = 360° / n
Number of Sides n = 360° / (Exterior Angle)

Figure 1.2: The Linear Pair Theorem and Exterior Angle Relationships.

4. Step-by-Step Curriculum Exemplars (from NaCCA Syllabus)

Exemplar 1: Irregular Hexagon Algebraic Angle Problem (PDF Page 34)

Problem: An irregular convex hexagon has interior angles: 120°, (5x - 6)°, (4x + 14)°, 7x°, (8x - 8)°, and 6x°. Find x and all six angles.



Theorem: Sum of interior angles of a convex hexagon = (6 - 2) * 180° = 720°

Worked Solution:

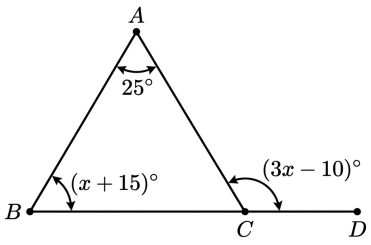
1. Theoretical sum for hexagon (n = 6):
   S = (6 - 2) × 180° = 4 × 180° = **720°**
2. Algebraic equation:
   120 + (5x - 6) + (4x + 14) + 7x + (8x - 8) + 6x = 720
3. Group variable and constant terms:
   (5x + 4x + 7x + 8x + 6x) + (120 - 6 + 14 - 8) = 720
   30x + 120 = 720 ⇒ 30x = 600 ⇒ **x = 20**
4. Individual angles:
   - 120°
   - 5(20) - 6 = **94°**
   - 4(20) + 14 = **94°**
   - 7(20) = **140°**
   - 8(20) - 8 = **152°**
   - 6(20) = **120°**

Verification: 120 + 94 + 94 + 140 + 152 + 120 = 720° (Exact match).

Figure 1.3: Geometric Schema for Exemplar 1 (Irregular Hexagon Angles).

Exemplar 2: Triangle Remote Exterior Angle Theorem (PDF Page 34)

Problem: In triangle ABC, angle A = 25°, angle B = (x + 15)°, and remote exterior angle ACD = (3x - 10)°. Solve for x and all angles.



Worked Solution:

1. **Theorem:** The exterior angle of a triangle equals the sum of its two remote interior angles.
   Angle ACD = Angle A + Angle B
2. Algebraic equation:
   3x - 10 = 25 + (x + 15)
   3x - 10 = x + 40
3. Solve for x:
   3x - x = 40 + 10 ⇒ 2x = 50 ⇒ **x = 25**
4. Calculate angle magnitudes:
   - Angle A = 25°
   - Angle B = 25 + 15 = **40°**
   - Exterior Angle ACD = 3(25) - 10 = **65°**
   - Interior Angle ACB = 180° - 65° = **115°**

Verification: 25° + 40° + 115° = 180° (Triangle sum confirmed).

Figure 1.4: Triangle Exterior Angle Relationship Schema.

5. Practice Exercises (40 Questions across 8 Sets)

Complete these 8 structured exercise sets to achieve master fluency in polygon geometry.

Exercise 1: Nomenclature & Diagonal Triangulation

1. State the formal geometric names of polygons with: (a) 7 sides, (b) 9 sides, (c) 10 sides, (d) 11 sides, (e) 12 sides.
2. How many diagonals can be drawn from a single vertex of: (a) a hexagon, (b) a nonagon, (c) a 16-gon?
3. How many non-overlapping triangles are formed by diagonals from one vertex of: (a) an octagon, (b) an undecagon, (c) a 20-gon?
4. A student counts 9 diagonals originating from a single vertex of a polygon. How many sides does this polygon have?
5. A polygon is partitioned into 11 triangles by diagonals from one vertex. State its number of sides and formal geometric name.

Exercise 2: Sum of Interior Angles of Polygons

1. Calculate the total sum of interior angles for: (a) a heptagon, (b) an octagon, (c) a decagon.
2. Determine the sum of the interior angles of a 14-sided polygon.
3. The sum of the interior angles of a polygon is 1980°. How many sides does it have?
4. Is it mathematically possible for a convex polygon to have an interior angle sum of 2400°? Justify rigorously.
5. Three angles of a pentagon are 100°, 110°, and 125°. If the remaining two angles are equal, find the size of each remaining angle.

Exercise 3: Regular Polygons — Interior & Exterior Angles

1. Find the measure of each interior angle and each exterior angle of a regular octagon.
2. Find the measure of each interior angle and each exterior angle of a regular decagon.
3. Calculate the size of each exterior angle of a regular dodecagon (n = 12).
4. What is the measure of each interior angle of a regular polygon with 18 sides?
5. The ratio of an interior angle to an exterior angle of a regular polygon is 5 : 1. Determine both angle values.

Exercise 4: Reverse Problems — Finding the Number of Sides (n)

1. Each exterior angle of a regular polygon measures 30°. How many sides does it have?
2. Each interior angle of a regular polygon is 140°. Determine its number of sides.
3. Each interior angle of a regular polygon is 165°. Calculate its number of sides.
4. The exterior angle of a regular polygon is 18°. What is the specific geometric name of this polygon?
5. The sum of the interior angles of a regular polygon is four times the sum of its exterior angles. Find n.

Exercise 5: Solving for Variables in Irregular Polygons

1. The angles of a quadrilateral are x°, (2x + 10)°, (3x - 20)°, and (4x + 10)°. Find x and the largest angle.
2. The interior angles of a pentagon are x°, (x + 20)°, (x + 40)°, (x + 60)°, and (x + 80)°. Determine x and all five angles.
3. An irregular hexagon has five angles measuring 110°, 125°, 115°, 130°, and 100°. Calculate the sixth angle.
4. The interior angles of a hexagon are 2x, 3x, 4x, 5x, 6x, and 7x. Find x and the smallest angle.
5. In a heptagon, six of the angles each measure 130°. Calculate the measure of the seventh angle.

Exercise 6: Triangles, Parallel Lines, and Exterior Angle Theorems

1. In triangle ABC, exterior angle at C is 132°. If angle A = 54°, calculate angle B.
2. The exterior angles of a triangle are in the ratio 2 : 3 : 4. Find all exterior and interior angles.
3. In a triangle, two remote interior angles are (2x - 5)° and (3x + 15)°, and the exterior angle is (6x - 20)°. Solve for x.
4. Two parallel lines are intersected by a transversal. If two co-interior angles are (3x + 15)° and (2x + 5)°, find x and both angles.
5. In quadrilateral ABCD with AB parallel to CD, angle A = 75° and angle B = 110°. Determine angles C and D.

Exercise 7: Real-World Architecture, Tiling, and Tessellation

1. Explain mathematically why regular hexagons can tessellate a flat surface on their own, while regular octagons cannot.
2. A hexagonal gazebo roof forms a regular hexagon. What angle do two adjacent boundary beams form?
3. A stop sign is a regular octagon. Calculate the interior angle between any two consecutive edges.
4. An architect combines regular octagons and squares to tile a floor. Prove that two octagons and one square meet seamlessly at a vertex.
5. A soccer ball pairs regular pentagons and hexagons. What is the difference between their interior angles?

Exercise 8: Challenge & Multi-Step Geometric Deduction

1. A polygon of n sides has two right angles, while the remaining (n - 2) angles each measure 150°. Determine n.
2. The ratio of the interior angle of regular polygon A to polygon B is 3 : 4. If polygon A has 6 sides, how many sides has B?
3. Five angles of a hexagon are each equal to x°, and the sixth angle is 80°. Calculate x.
4. In a convex decagon, nine interior angles each measure 145°. What is the tenth angle? Is the decagon convex?
5. The difference between an interior angle of an (n + 1)-gon and an n-gon is 4°. Find n.

6. Complete Step-by-Step Worked Solutions

Full mathematical derivations and checks for all 40 practice problems.

Solutions to Exercise 1

1. (a) Heptagon, (b) Nonagon, (c) Decagon, (d) Undecagon, (e) Dodecagon.
2. Diagonals from one vertex = n - 3: (a) 6 - 3 = 3, (b) 9 - 3 = 6, (c) 16 - 3 = 13.
3. Triangles = n - 2: (a) 8 - 2 = 6, (b) 11 - 2 = 9, (c) 20 - 2 = 18.
4. n - 3 = 9 $\Rightarrow$ n = 12 sides (Dodecagon).
5. n - 2 = 11 $\Rightarrow$ n = 13 sides (Tridecagon / 13-gon).

Solutions to Exercise 2

1. S = (n - 2) × 180°: (a) Heptagon = 5 × 180° = 900°, (b) Octagon = 6 × 180° = 1080°, (c) Decagon = 8 × 180° = 1440°.
2. n = 14 $\Rightarrow$ S = 12 × 180° = 2160°.
3. (n - 2) × 180 = 1980 $\Rightarrow$ n - 2 = 11 $\Rightarrow$ n = 13 sides.
4. (n - 2) × 180 = 2400 $\Rightarrow$ n - 2 = 13.333. Since n must be an integer, 2400° is impossible.
5. Pentagon sum = 540°. Sum known = 100 + 110 + 125 = 335°. Remaining = 540 - 335 = 205°. Each angle = 205° / 2 = 102.5°.

Solutions to Exercise 3

1. Octagon (n = 8): Exterior = 360° / 8 = 45°; Interior = 180° - 45° = 135°.
2. Decagon (n = 10): Exterior = 360° / 10 = 36°; Interior = 180° - 36° = 144°.
3. Dodecagon (n = 12): Exterior = 360° / 12 = 30°.
4. n = 18: Exterior = 360° / 18 = 20°; Interior = 180° - 20° = 160°.
5. x + 5x = 180° $\Rightarrow$ 6x = 180° $\Rightarrow$ x = 30°. Exterior = 30°, Interior = 150°.

Solutions to Exercise 4

1. n = 360° / 30° = 12 sides.
2. Exterior = 180° - 140° = 40°. n = 360° / 40° = 9 sides.
3. Exterior = 180° - 165° = 15°. n = 360° / 15° = 24 sides.
4. n = 360° / 18° = 20 sides (Icosagon or 20-gon).
5. (n - 2) × 180 = 4 × 360 = 1440 $\Rightarrow$ n - 2 = 8 $\Rightarrow$ n = 10 sides (Decagon).

Solutions to Exercise 5

1. $x + (2x + 10) + (3x - 20) + (4x + 10) = 360 \Rightarrow 10x = 360 \Rightarrow x = 36$. Angles: 36°, 82°, 88°, 154°. Largest = 154°.
2. $5x + 200 = 540 \Rightarrow 5x = 340 \Rightarrow x = 68$. Angles: 68°, 88°, 108°, 128°, 148°.
3. Hexagon sum = 720°. Known sum = 110 + 125 + 115 + 130 + 100 = 580°. Sixth angle = 720 - 580 = 140°.
4. $27x = 720 \Rightarrow x = 80/3 = 26.67°$. Smallest angle = $2x = 160/3 = 53.33°$ (53° 20').
5. Heptagon sum = 900°. Six angles = 6 × 130° = 780°. Seventh angle = 900 - 780 = 120°.

Solutions to Exercise 6

1. Exterior angle = sum of remote interior: 54° + B = 132° ⇒ B = 132 - 54 = 78°.
2. Exterior ratio parts = 2 + 3 + 4 = 9. 1 part = 360° / 9 = 40°. Exterior: 80°, 120°, 160°. Interior: 100°, 60°, 20°.
3. $(2x - 5) + (3x + 15) = 6x - 20 \Rightarrow 5x + 10 = 6x - 20 \Rightarrow x = 30$.
4. Co-interior sum = 180°: $(3x + 15) + (2x + 5) = 180 \Rightarrow 5x + 20 = 180 \Rightarrow 5x = 160 \Rightarrow x = 32$. Angles: 111° and 69°.
5. Co-interior: Angle D = 180 - 75 = 105°; Angle C = 180 - 110 = 70°.

Solutions to Exercise 7

1. Hexagon interior = 120°. 360° / 120° = 3 (exact integer), tiles cleanly around a vertex. Octagon interior = 135°. 360 / 135 = 2.67 (not integer), cannot tile alone.
2. Regular hexagon interior angle = (6 - 2) × 180° / 6 = 120°.
3. Regular octagon interior angle = (8 - 2) × 180° / 8 = 135°.
4. Two octagons = 2 × 135° = 270°. One square = 90°. Total = 270° + 90° = 360° (complete tile vertex with 0 gap).
5. Hexagon interior = 120°, Pentagon interior = 108°. Difference = 120 - 108 = 12°.

Solutions to Exercise 8

1. $2(90) + (n - 2)150 = (n - 2)180 \Rightarrow 180 + 150n - 300 = 180n - 360 \Rightarrow 30n = 240 \Rightarrow n = 8$ sides (Octagon).
2. Polygon A (n = 6) interior = 120°. Ratio A:B = 3:4 ⇒ Angle B = 160°. Ext angle B = 20°. Sides of B = 360° / 20° = 18 sides.
3. $5x + 80 = 720 \Rightarrow 5x = 640 \Rightarrow x = 128°$.
4. Decagon sum = 1440°. 9 × 145 = 1305°. Tenth angle = 1440 - 1305 = 135°. Yes, convex since all angles < 180°.
5. $360/n - 360/(n+1) = 4 \Rightarrow 90/n - 90/(n+1) = 1 \Rightarrow n(n+1) = 90 \Rightarrow n^2 + n - 90 = 0 \Rightarrow (n+10)(n-9) = 0 \Rightarrow n = 9$ sides.